Mengxiao (Jackson) Wang

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Research Interests

Security of LLM agents that act in high-stakes settings, with a focus on LLM trading agents: prompt injection, robustness, and controls that hold against adaptive attackers.

Education

Texas A&M University — College Station, TX
Ph.D. in Computer Science. Advisor: Nitesh Saxena. GPA 4.0/4.0 — Aug 2023 – present

Carnegie Mellon University — Pittsburgh, PA
M.S. in Electrical and Computer Engineering. GPA 3.78/4.0 — Sep 2021 – May 2023

University of Minnesota, Twin Cities — Minneapolis, MN
B.A. in Computer Science, Minor in Statistics. GPA 3.815/4.0 — Jan 2018 – May 2021

Publications

Peer-reviewed

[C1] **Mengxiao Wang**, Jianwei Huang, Guofei Gu. Demystifying Progressive Web Application Permission Systems. *IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2026.

[C2] Jianwei Huang, Sridatta Chintapalli, **Mengxiao Wang**, Guofei Gu. Beyond Visual Confusion: Understanding How Inconsistencies in ENS Normalization Facilitate Homoglyph Attacks. *The ACM Web Conference (WWW)*, 2025.

[J1] **Mengxiao Wang**. Application of Computer Vision in Logistics Warehouse Safety Management. *Software*, 41(10):180–183, 2024.

Preprints

[P1] **Mengxiao Wang**, Nitesh Saxena. SoK: Trading Agents or Market Crashers? Dissecting Robustness and Security Failures in Academic Financial LLM Trading Schemes. arXiv:2609.19705, 2026.

[P2] **Mengxiao Wang**, Yuxuan Zhang, Guofei Gu. PromptSleuth: Detecting Prompt Injection via Semantic Intent Invariance. arXiv:2508.20890, 2025.

[P3] **Mengxiao Wang**, Guofei Gu. GuardianPWA: Enhancing Security Throughout the Progressive Web App Installation Lifecycle. arXiv:2509.13561, 2025.

[P4] **Mengxiao Wang**. AttackEval: A Systematic Empirical Study of Prompt Injection Attack Effectiveness Against Large Language Models. arXiv:2604.03598, 2026. (Posted under the name Jackson Wang.)

Research Experience

SPIES Lab, Texas A&M University — College Station, TX
Graduate Research Assistant, advised by Nitesh Saxena — 2025 – present

- Systematized robustness and security failures across 15 academic LLM trading-agent designs [P1].

SUCCESS Lab, Texas A&M University — College Station, TX
Graduate Research Assistant, advised by Guofei Gu — Aug 2023 – 2025

- Measured permission enforcement in Progressive Web Apps across browsers and platforms [C1, P3] and built PromptSleuth, a prompt-injection detector based on task-level intent [P2].

CyLab, Carnegie Mellon University — Pittsburgh, PA
Research Assistant — Jul 2021 – May 2023

- Extended a taint analysis to TypeScript for a crawl of 2.1 million npm packages, raising its detection of command-injection flaws by 40%. The work contributed to more than 30 CVEs reported by the team.

Inspur Jinan, China
LLM Security Research Intern May 2024 – Jul 2024

- Built a fuzzing harness for LLM applications (1,000+ test cases) and found three classes of content-restriction bypass. Added semantic analysis to an input-output filter, cutting its false-positive rate from 22.3% to 4%.

Vulnerability Disclosures

- Progressive Web App permission and installation flaws [C1]: reported 12 issues to Google Chrome (7) and Mozilla Firefox (5), and further issues to Samsung Internet. An Android Chrome crash I reported, which locked users out of the browser, was confirmed by Google and is fixed in Chrome 131; Mozilla's security team flagged two reports as privacy issues (sec-want) and adopted one as its tracking bug for isolating installed web apps (Bugzilla 1868204).
- Prompt-injection patterns found with PromptSleuth [P2]: disclosed to OpenAI.
- npm packages: more than 30 CVEs for code-execution and command-injection flaws, as a member of the CMU CyLab team.

Grants and Awards

OpenAI Researcher Access Program Grant 2025

Teaching, Mentoring, and Service

Teaching Assistant, Texas A&M: CSCE 765 Network Security (Spring 2026); CSCE 689/489 Security Engineering (Fall 2025)
Teaching Assistant, CMU: 18-631 Introduction to Information Security (Fall 2022); 18-613 Foundations of Computer Systems (Spring 2022); 17-683 Data Structures for Application Programmers (Spring 2023); 17-681 Java for Application Programmers (Spring 2023)
Mentoring: one undergraduate student at Texas A&M, on trading-agent security experiments (Spring 2026 – present)
Program Committee: The ACM Web Conference (WWW) 2026, Security and Privacy track
Artifact Evaluation Committee: NDSS 2027
External Reviewer: NDSS 2025
Sub-reviewer: USENIX Security 2024; ACM CCS 2025; NDSS 2026

Skills

Programming: Python, C/C++, Go, Rust, TypeScript/JavaScript, SQL
Methods: taint analysis, LLM evaluation harnesses, bootstrap confidence intervals, paired tests (McNemar), Firth regression, event studies